IKEA Inventory Management System | INDIA

-SQL Project | MySQL | Inventory Analytics



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Project Overview

Project Objective

Develop a **relational database management system** for IKEA to analyse inventory, customer purchases, store performance, supplier performance, and sales trends using **MySQL**.

Dataset Description

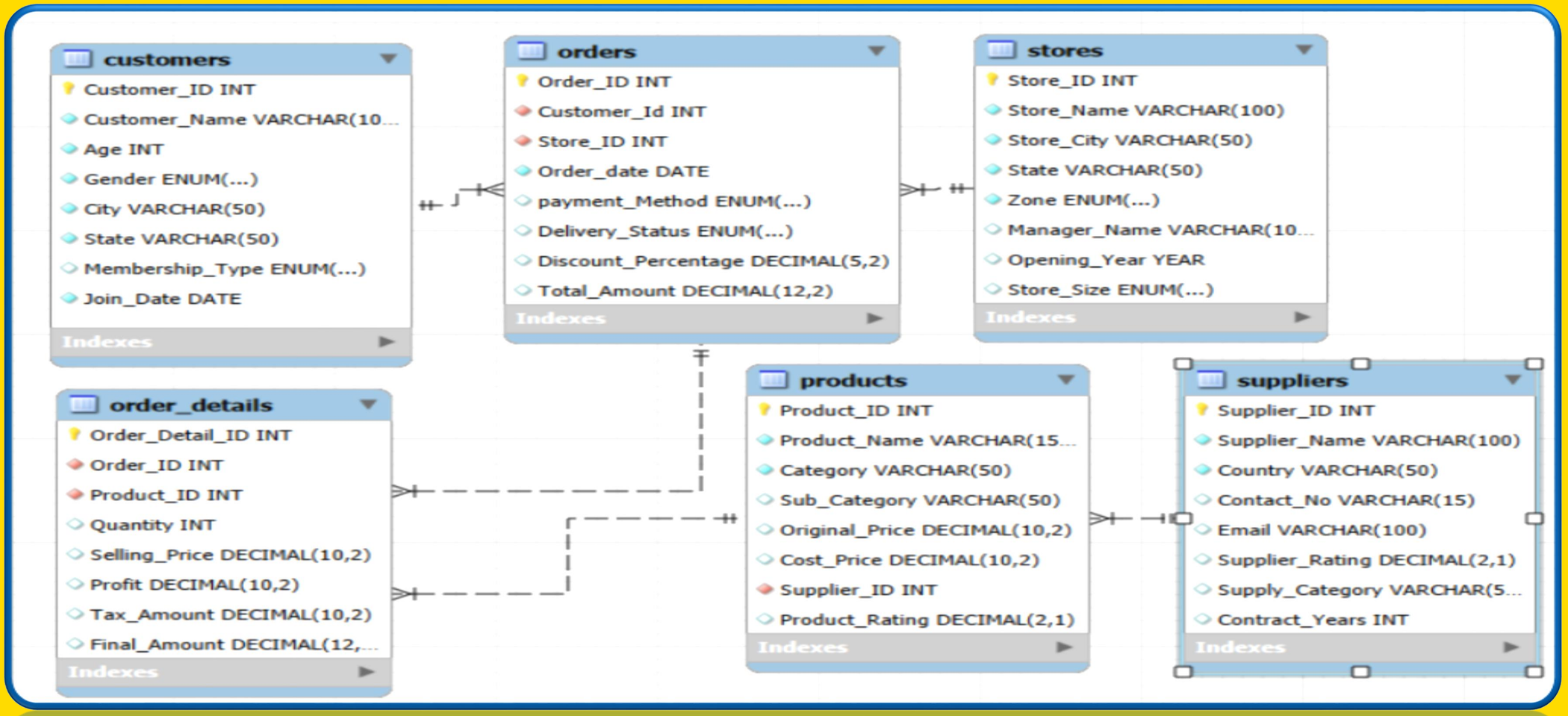
The project uses a custom **IKEA Inventory Management dataset** containing information about customers, products, suppliers, stores, orders, and order details. The database is designed to simulate **real-world retail operations** and support **business reporting**.

SQL Concepts Covered

- ✔ **Database Design**
- ✔ **Table Creation**
- ✔ **Constraints**
- ✔ **Joins**
- ✔ **Aggregate Functions**
- ✔ **Subqueries**
- ✔ **CASE**
- ✔ **Common Table Expressions (CTE)**
- ✔ **Window Functions**
- ✔ **Business Reporting**

DATABASE SCHEMAS

ER DIAGRAM



Display each store's total orders, total revenue, average order value, and total profit.




```sql
select s.Store_id, s.Store_Name, s.Store_City,
count(distinct o.order_id) as Total_Orders,
round(sum(od.Final_amount),2) as Total_Revenue,
round(avg(o.Total_Amount), 2) as Average_Order_Value,
round(sum(od.profit), 2) as Total_Profit
from stores s inner join orders o on
s.store_id = o.store_id
inner join order_details od on o.order_id = od.order_id
where delivery_status = 'Delivered'
group by s.Store_id, s.Store_Name, s.Store_City
order by Total_Revenue desc;
```

Store_id	Store_Name	Store_City	Total_Orders	Total_Revenue	Average_Order_Value	Total_Profit
4	IKEA Mumbai City	Mumbai	16	572034.50	37269.36	131020.00
1	IKEA Bengaluru	Bengaluru	18	514550.80	28936.99	119440.00
5	IKEA Delhi NCR	Gurugram	15	430898.24	23629.43	98940.00
9	IKEA Kolkata	Kolkata	12	405430.30	24837.40	93200.00
6	IKEA Chennai	Chennai	17	363410.50	17683.45	85170.00
2	IKEA Hyderabad	Hyderabad	18	359742.16	21965.23	86200.00
3	IKEA Navi Mumbai	Navi Mumbai	14	315848.24	18069.10	73770.00
7	IKEA Pune	Pune	10	222162.14	16554.15	53170.00
10	IKEA Lucknow	Lucknow	10	218730.70	21818.92	52070.00
8	IKEA Ahmedabad	Ahmedabad	11	129978.18	10460.04	32280.00

Classify Top 10 Customers based on Total Spending

```sql
select c.Customer_ID, c.Customer_Name,
count(distinct o.Order_id) As Total_Orders,
ROund(sum(o.Total_Amount),2) as Total_Spending,
case
when sum(o.Total_Amount) > 150000 then 'Platinum'
when sum(o.Total_Amount) between 80000 and 150000 then 'Gold'
when sum(o.Total_Amount) between 30000 and 79000 then 'Silver'
else 'Bronze'
end as 'Customer_Segment'
from Customers c inner join orders o on c.customer_id = o.customer_id
where delivery_Status = 'delivered'
group by c.Customer_ID, c.Customer_Name
order by Total_Spending desc limit 10;
```





Customer_ID	Customer_Name	Total_Orders	Total_Spending	Customer_Segment
188	Preeti Khanna	2	90856.46	Gold
183	Tarun Verma	2	82594.10	Gold
145	Vikas Gupta	2	77297.87	Silver
189	Kunal Sharma	1	73156.46	Silver
105	Sandeep Kumar	1	70798.82	Silver
142	Ishita Arora	1	70798.82	Silver
158	Ritika Kapoor	1	70798.82	Silver
167	Ashish Saxena	3	70517.02	Silver
73	Ananya Nair	4	68535.79	Silver
140	Bhavna Nair	2	67397.92	Silver

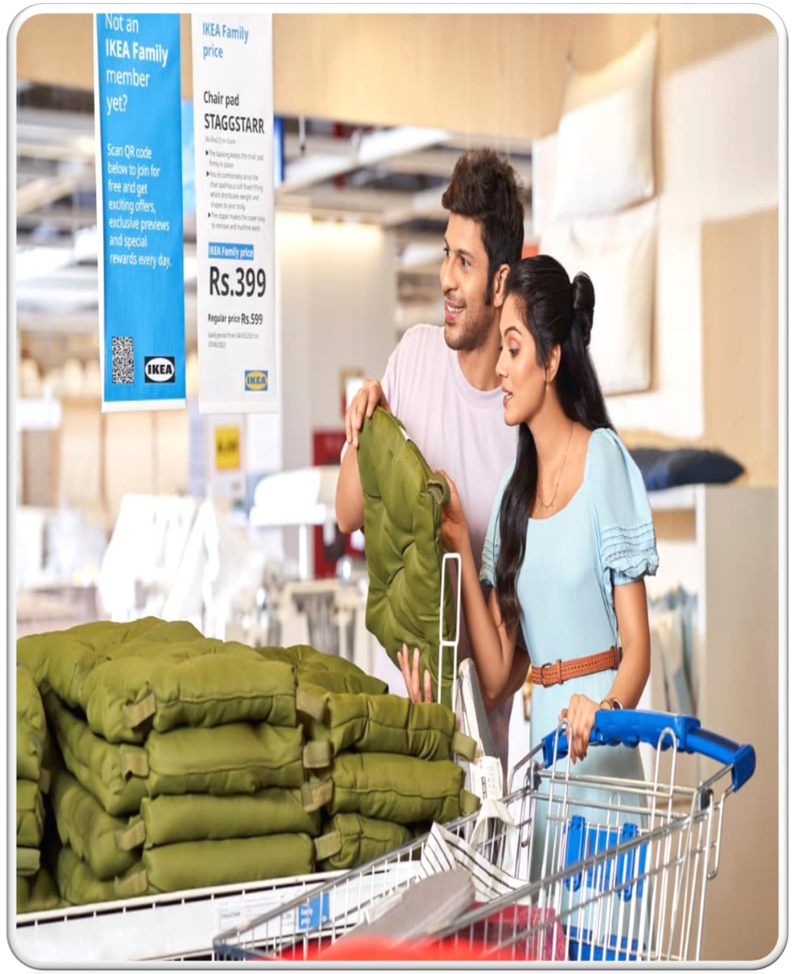
Find the highest revenue-generating product in each product category.
If two products have the same revenue, show both.




```sql
with Product_Sales as
(select p.Product_ID,p.Product_Name,
p.category as Product_Category,
sum(od.Quantity) as Total_Quantity_Sold,
round(sum(od.Final_Amount),2) as Total_Revenue
from Products p inner join Order_Details od
on p.product_id = od.product_id
inner join Orders o on od.order_id = o.order_id
where o.delivery_status = 'Delivered'
group by p.Product_id, p.Product_Name, p.Category
),
Product_Rank as (
select *, dense_rank()
over(partition by Product_Category order by Total_Revenue desc)
as Product_Rank from Product_Sales)
select Product_ID, Product_Name,
Product_Category, Total_Quantity_Sold, Total_Revenue, Product_Rank
from Product_Rank where Product_Rank = 1 order by Product_Category;
```

Product_ID	Product_Name	Product_Category	Total_Quantity_Sold	Total_Revenue	Product_Rank
81	GODMORGON Vanity	Bathroom	7	165191.74	1
101	VESTERÖY Mattress Queen	Bedroom	10	271388.20	1
42	HOVET Mirror	Decor	6	63712.92	1
61	INGATORP Dining Table	Dining	8	212390.56	1
157	IDANÄS Wardrobe	Furniture	7	289091.74	1
47	TROFAST Storage Unit	Kids	3	24776.46	1
176	OFTAST Dinner Plate	Kitchen	49	11511.18	1
31	HEKTAR Floor Lamp	Lighting	5	41294.10	1
30	FRIHETEN Sofa Bed	Living Room	11	778787.02	1
18	MARKUS Office Chair	Office	6	134512.92	1
71	ÄPPLARÖ Garden Table	Outdoor	10	188788.20	1
91	TRÅDFRI LED Bulb	Smart Home	18	19094.76	1
54	IVAR Shelf Unit	Storage	2	18877.64	1

Find Top 10 Customers whose total purchase amount is greater than the average spending of all customers.




```sql
select c.Customer_ID, c.Customer_Name,
count(distinct o.order_id) as Total_Orders,
Round(sum(o.Total_Amount), 2) as Customer_Spending from customers c
inner join orders o on c.customer_id= o.customer_id
where o.delivery_status = 'delivered'
group by c.Customer_ID, c.Customer_Name
having sum(o.Total_Amount) > (select avg(Customer_Spending)
from (select Customer_ID, sum(Total_Amount) as Customer_Spending
from orders where delivery_status = 'delivered'
group by Customer_ID) as Avg_Spending)
order by Customer_Spending desc limit 10;
```

Customer_ID	Customer_Name	Total_Orders	Customer_Spending
188	Preeti Khanna	2	90856.46
183	Tarun Verma	2	82594.10
145	Vikas Gupta	2	77297.87
189	Kunal Sharma	1	73156.46
105	Sandeep Kumar	1	70798.82
158	Ritika Kapoor	1	70798.82
142	Ishita Arora	1	70798.82
167	Ashish Saxena	3	70517.02
73	Ananya Nair	4	68535.79
140	Bhavna Nair	2	67397.92

Display monthly revenue. Also show Previous month's revenue, Revenue Difference and Growth %.




```sql
WITH Monthly_Sales AS
(
SELECT YEAR(Order_Date) AS Year, MONTH(Order_Date) AS Month,
ROUND(SUM(Total_Amount),2) AS Total_Revenue FROM Orders
WHERE Delivery_Status = 'Delivered'
GROUP BY YEAR(Order_Date), MONTH(Order_Date)
)
SELECT Year, Month, Total_Revenue, LAG(Total_Revenue)
OVER( ORDER BY Year, Month) AS Previous_Month_Revenue,
ROUND( Total_Revenue - LAG(Total_Revenue)
OVER( ORDER BY Year, Month),2) AS Revenue_Difference,
ROUND(((Total_Revenue - LAG(Total_Revenue)
OVER(ORDER BY Year, Month)) / LAG(Total_Revenue)
OVER( ORDER BY Year, Month)
 ) * 100,2 ) AS Growth_Percentage
FROM Monthly_Sales ORDER BY Year, Month;
```

Year	Month	Total_Revenue	Previous_Month_Revenue	Revenue_Difference	Growth_Percentage
2025	1	152569.51	NULL	NULL	NULL
2025	2	204436.51	152569.51	51867.00	34.00
2025	3	296133.97	204436.51	91697.46	44.85
2025	4	505236.51	296133.97	209102.54	70.61
2025	5	469593.98	505236.51	-35642.53	-7.05
2025	6	454622.14	469593.98	-14971.84	-3.19
2025	7	601765.78	454622.14	147143.64	32.37
2025	8	395042.76	601765.78	-206723.02	-34.35

Rank Top 10 products based on total revenue.



```sql
select p.product_id,p.Product_Name, p.Category, sum(od.Quantity) as Total_Quantity_Sold,
round(sum(od.Final_Amount),2) as Total_Revenue,
row_number() over(order by sum(od.final_amount) desc) as Row_Num,
rank() over(order by sum(od.final_amount) desc) as Product_Rank,
dense_rank() over(order by sum(od.final_amount) desc) as Dense_Product_Rank
from products p inner join order_details od on p.product_id = od.product_id
inner join orders o on od.order_id = o.order_id
where delivery_Status='delivered' group by p.product_id,p.Product_Name, p.Category
order by Total_Revenue limit 10;
```

product_id	Product_Name	Category	Total_Quantity_Sold	Total_Revenue	Row_Num	Product_Rank	Dense_Product_Rank
38	365+ Mug	Kitchen	6	1408.92	46	46	46
170	RISATORP Basket	Storage	1	1532.82	45	45	45
36	365+ Dinner Plate	Kitchen	4	1883.28	44	44	44
179	KALAS Kids Bowl Set	Kitchen	6	2116.92	43	43	43
172	KVISSLE Magazine File	Office	2	2357.64	42	42	42
178	POKAL Glass	Kitchen	24	2803.68	41	41	41
180	KALAS Cutlery Set	Kitchen	6	2824.92	40	40	40
165	GRADVIS Vase	Decor	2	2829.64	39	39	39
174	ROMMA Monitor Stand	Office	1	2948.82	38	38	38
161	FISKBO Photo Frame	Decor	6	3532.92	37	37	37

For every supplier display Top 10 Total Products, Total Orders, Total Revenue Generated and Average Product Rating.



```sql
select s.Supplier_ID, s.supplier_name,
count(distinct p.Product_ID) as Total_Products,
round( avg(p.Product_Rating ),2) as average_rating,
count(distinct od.order_ID) as total_orders,
round(sum(od.Final_Amount),2) as Total_Revenue
from  products p inner join Suppliers s on p.supplier_id = s.supplier_id
left join  order_details od on p.product_id = od.product_id
left join orders o on o.order_id=od.order_id
where o.delivery_status = 'Delivered'
group by s.Supplier_ID, s.supplier_name order by Total_Revenue desc limit 10;
```

Supplier_ID	supplier_name	Total_Products	average_rating	total_orders	Total_Revenue
5	Elegant Home Solutions	3	4.83	20	1012416.40
2	Nordic Furnishings AB	3	4.90	13	625384.66
22	Classic Woodcraft	3	4.85	11	385844.66
4	Comfort Living Ltd	2	4.80	12	349265.84
23	Elite Manufacturing Ltd	5	4.71	18	239503.42
15	Urban Office Systems	5	4.81	15	194080.50
1	Scandinavian Wood Pvt Ltd	2	4.80	8	193510.56
19	Outdoor Living Pvt Ltd	1	4.70	8	188788.20
8	Modern Storage Pvt Ltd	3	4.75	8	77865.84
14	Royal Décor House	1	4.80	6	63712.92

Highest Product Rating from each Product Category.



```
SELECT product_name, category, sub_category, product_rating
FROM (SELECT product_name, category as Product_Category, sub_category, product_rating,
ROW_NUMBER() OVER (PARTITION BY category ORDER BY product_rating DESC) AS rn
FROM products) as p
WHERE rn = 1 order by product_rating desc;
```

product_name	category	sub_category	product_rating
VESTERÖY Mattress King	Bedroom	Mattresses	4.9
STOCKHOLM Rug	Decor	Rugs	4.9
MÖRBYLÅNGA Table	Dining	Tables	4.9
PAX Wardrobe 150cm	Furniture	Wardrobes	4.9
MAJGULL Blackout Curtains	Home Textiles	Curtains	4.9
LILLABO Train Set	Kids	Toys	4.9
RÅSKOG Utility Trolley	Kitchen	Storage	4.9
KIVIK 3 Seat Sofa	Living Room	Sofas	4.9
MARKUS Office Chair	Office	Office Chairs	4.9
SYMFONISK Lamp Speaker	Smart Home	Speakers	4.9
GODMORGON Vanity	Bathroom	Vanities	4.8
HEMNES Shoe Cabinet	Entrance	Shoe Storage	4.8
MULIG Clothes Rack	Home Organi...	Wardrobe	4.8
HEKTAR Floor Lamp	Lighting	Floor Lamps	4.8
SJÄLLAND Outdoor Table	Outdoor	Tables	4.8
LURVIG Pet Bed	Pet Products	Pet Beds	4.8
STRÅLA LED Star	Seasonal	Lighting	4.8
PANSARTAX Storage Box	Storage	Boxes	4.8

Calculate Store Revenue and Percentage contribution to Total Revenue.




```sql
with Highest_Contribution as
(
select  s.Store_ID, s.Store_Name , s.Store_City,
round(sum(od.Final_Amount),2) as Total_Revenue from Stores s
inner join orders o on s.store_id = o.store_id
inner join order_Details od on o.Order_id = od.order_id
where o.delivery_status = 'delivered'
group by s.Store_ID, s.Store_Name , s.Store_City
)
select Store_ID, Store_Name, Store_City, Total_Revenue,
round((Total_Revenue/ sum(Total_Revenue) over()) * 100,2) as Revenue_Contribution_Percentage
from Highest_Contribution
order by Total_Revenue desc;
```

Store_ID	Store_Name	Store_City	Total_Revenue	Revenue_Contribution_Percentage
4	IKEA Mumbai City	Mumbai	572034.50	16.19
1	IKEA Bengaluru	Bengaluru	514550.80	14.57
5	IKEA Delhi NCR	Gurugram	430898.24	12.20
9	IKEA Kolkata	Kolkata	405430.30	11.48
6	IKEA Chennai	Chennai	363410.50	10.29
2	IKEA Hyderabad	Hyderabad	359742.16	10.18
3	IKEA Navi Mumbai	Navi Mumbai	315848.24	8.94
7	IKEA Pune	Pune	222162.14	6.29
10	IKEA Lucknow	Lucknow	218730.70	6.19
8	IKEA Ahmedabad	Ahmedabad	129978.18	3.68

Generate a Business report showing Customer Name, Store Name, Product Category, Total Orders, Revenue, Profit, Customer Segment and Product Rank within Category which Only include Delivered Orders and having Revenue > ₹20,000.




```sql
with Customer_Report as (
select c.Customer_ID, c.Customer_Name, s.Store_Name, p.Category,
count(distinct o.order_id) as Total_Orders,
round(sum(od.Final_Amount),2) as Total_Revenue,
Round(sum(od.profit),2) as Total_Profit,
case when sum(od.Final_Amount) > 150000 then 'Platinum'
when sum(od.Final_Amount) between 80000 and 150000 then 'Gold'
when sum(od.Final_Amount) between 30000 and 79999 then 'Silver'
else 'Bronze'
end as Customer_Segment
from customers c inner join orders o on c.customer_id = o.Customer_id
inner join Stores s on o.store_id = s.store_id
inner join Order_Details od on o.Order_ID = od.Order_ID
inner join Products p on od.Product_ID = p.Product_ID
where o.delivery_status = 'delivered'
group by c.Customer_ID, c.Customer_Name, s.Store_Name, p.Category
having sum(od.Final_Amount) > 20000)
select Customer_Name, Store_Name, Category, Total_Orders,
Total_Revenue, Total_Profit, Customer_Segment,
dense_rank() over(partition by Category
order by Total_Revenue desc) as Product_Category_Rank
from Customer_Report
order by Total_Revenue desc;
```

Customer_Name	Store_Name	Category	Total_Orders	Total_Revenue	Total_Profit	Customer_Segment	Product_Category_Rank
Tarun Verma	IKEA Mumbai City	Living Room	1	70798.82	16000.00	Silver	1
Sandeep Kumar	IKEA Bengaluru	Living Room	1	70798.82	16000.00	Silver	1
Karan Mehta	IKEA Navi Mumbai	Living Room	1	70798.82	16000.00	Silver	1
Kunal Sharma	IKEA Bengaluru	Living Room	1	70798.82	16000.00	Silver	1
Ritika Kapoor	IKEA Mumbai City	Living Room	1	70798.82	16000.00	Silver	1
Shreya Ghosh	IKEA Chennai	Living Room	1	70798.82	16000.00	Silver	1
Deepak Pandey	IKEA Pune	Living Room	1	70798.82	16000.00	Silver	1
Vikas Gupta	IKEA Delhi NCR	Living Room	1	70798.82	16000.00	Silver	1
Ishita Arora	IKEA Kolkata	Living Room	1	70798.82	16000.00	Silver	1
Bhavna Nair	IKEA Kolkata	Living Room	1	70798.82	16000.00	Silver	1
Abhishek Nair	IKEA Navi Mumbai	Living Room	1	70798.82	16000.00	Silver	1
Kajal Das	IKEA Kolkata	Furniture	1	64898.82	14000.00	Silver	1
Pallavi Ghosh	IKEA Lucknow	Living Room	1	58998.82	13000.00	Silver	2
Neha Bansal	IKEA Bengaluru	Living Room	1	58998.82	13000.00	Silver	2
Payal Mishra	IKEA Mumbai City	Furniture	1	54278.82	12000.00	Silver	2
Megha Jain	IKEA Delhi NCR	Furniture	1	54278.82	12000.00	Silver	2
Shreya Ghosh	IKEA Mumbai City	Furniture	1	54278.82	12000.00	Silver	2
Akash Malhotra	IKEA Delhi NCR	Furniture	1	54278.82	12000.00	Silver	2
Ananya Nair	IKEA Bengaluru	Furniture	1	54278.82	12000.00	Silver	2
Yash Agarwal	IKEA Delhi NCR	Dining	1	53098.82	12000.00	Silver	1
Preeti Khanna	IKEA Mumbai City	Dining	1	53098.82	12000.00	Silver	1
Aditi Kapoor	IKEA Pune	Furniture	1	41298.82	9500.00	Silver	3
Bhavna Nair	IKEA Kolkata	Furniture	1	41298.82	9500.00	Silver	3
Abhishek Roy	IKEA Mumbai City	Furniture	1	41298.82	9500.00	Silver	3

BUSINESS INSIGHTS



Store Performance

- ***Bangalore*** *Store generated the highest total revenue among all stores.*
- ***Hyderabad*** *Store recorded the highest number of customer orders.*
- *Some stores showed high order volume but comparatively lower average order value, indicating opportunities to improve product mix.*

Product Performance

- ***Wardrobe*** *and* ***Sofa*** *categories contributed the highest share of overall revenue.*
- *A few products remained unsold, highlighting potential inventory optimisation opportunities.*
- *Top-selling products consistently generated the majority of total sales.*

Supplier Performance

- *A small number of suppliers contributed most of the high-performing products.*
- *Suppliers with consistently higher product ratings also generated better revenue.*
- *Supplier performance can be evaluated using revenue, product ratings, and order volume.*

Sales Trends

- *Monthly revenue showed noticeable fluctuations throughout the year.*
- *Revenue contribution analysis identified the top-performing stores.*
- *Growth percentage analysis helps management monitor business performance over time.*

Customer Insights

- *A limited number of* ***high-value (Platinum)*** *customers contributed a significant portion of total revenue.*
- *Most customers belonged to the* ***Silver*** *segment, indicating potential for upselling.*
- *Customer segmentation helps create targeted marketing campaigns.*

Life at Home | Observations



Current and future home

63% of Indians feel positive about their current life at home, and 71% have a positive outlook on their future in the next two years

For 50% of Indians, home is their favourite place to be.

Their main concerns around life at home are their physical health (34%), the health of the people around them (30%), and their mental health (29%).

The most important aspect of an ideal home is that it helps them be physically or mentally stronger (34%). When they imagine their life in an ideal home, the top outcomes are feeling happy, relaxed and content, and confident.



Needs of home

Each emotional need is important to at least 65% of Indians, but security (74%), comfort (74%), and enjoyment (73%) rank highest

The needs which are *often* fulfilled at home for the largest shares of Indians are comfort (58%), security (56%), and belonging (53%).

Aspirations and accomplishment are the needs for which the gaps between importance and fulfilment are the largest. Those needs have the largest shares of Indians who find the respective needs important but never or rarely experience fulfilment at home.



Moving

56% of Indians are planning to move in the next two years, and an additional 18% are planning to stay in their current home but renovate

The likelihood of moving is highest for Generation Z and decreases somewhat with age. The older generations are more inclined than Generation Z to renovate. Homeowners are less likely than renters to plan to move, but more likely to renovate.

Among those planning to move, the main reasons are getting closer to something/someone (22%), getting more space (20%), and getting away from current area (15%). 18% want to get a more sustainable home.



Sleep

The number one key element in Indians' habits for best sleep is their favourite pillow (31%)

It is of similar importance to Indians as to the global average (28%), and slightly more important to women than to men (33%, 28%)

Other key elements for which IKEA products can be of help are complete darkness which is mentioned by a total of 26%, a night light (16%), and a heavy cover (9%).

SUMMARY



The **IKEA Inventory Management System** project demonstrates the practical application of SQL in solving real-world retail business problems through effective database design and data analysis. A relational database was developed using multiple interconnected tables to manage customers, products, suppliers, stores, orders, and order details. Advanced SQL concepts such as **Joins, Aggregate Functions, CASE Statements, Subqueries, Common Table Expressions (CTEs), and Window Functions** were implemented to generate meaningful business reports and performance insights. The analysis helped identify top-performing stores, best-selling product categories, supplier performance, customer purchasing behaviour, monthly sales trends, and unsold inventory. This project strengthened my understanding of relational database management, query optimization, and business intelligence reporting while demonstrating how SQL can transform raw transactional data into actionable insights for better business decision-making.

This project reflects my ability to design databases, write advanced SQL queries, analyse business data, and present insights in a structured and professional manner, making it a strong demonstration of my Data Analyst skills.

THANK YOU

"Turning Data into Meaningful Business Insights."

Thank you for reviewing my SQL project.
I look forward to applying my analytical skills to solve real-world business challenges and contribute as a Data Analyst.



Presented By -
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